ML Programming Assignment - Occupancy Detection

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1 Problem

Knowing whether a room is occupied is useful for tasks such as energy-saving HVAC control or smart lighting, and can be inferred from ambient sensor readings instead of privacy-sensitive means like cameras. We work with a dataset of 8143 records, each describing a room at a given moment through five sensor-derived quantities and a binary occupancy label:

1. Temperature, in Celsius degrees;
2. Humidity, in percentage;
3. Light, in Lux;
4. CO2, in ppm;
5. Humidity ratio, calculated using the measured temperature and relative humidity;
6. Occupancy: presence (1) or absence (0) of people in the room.

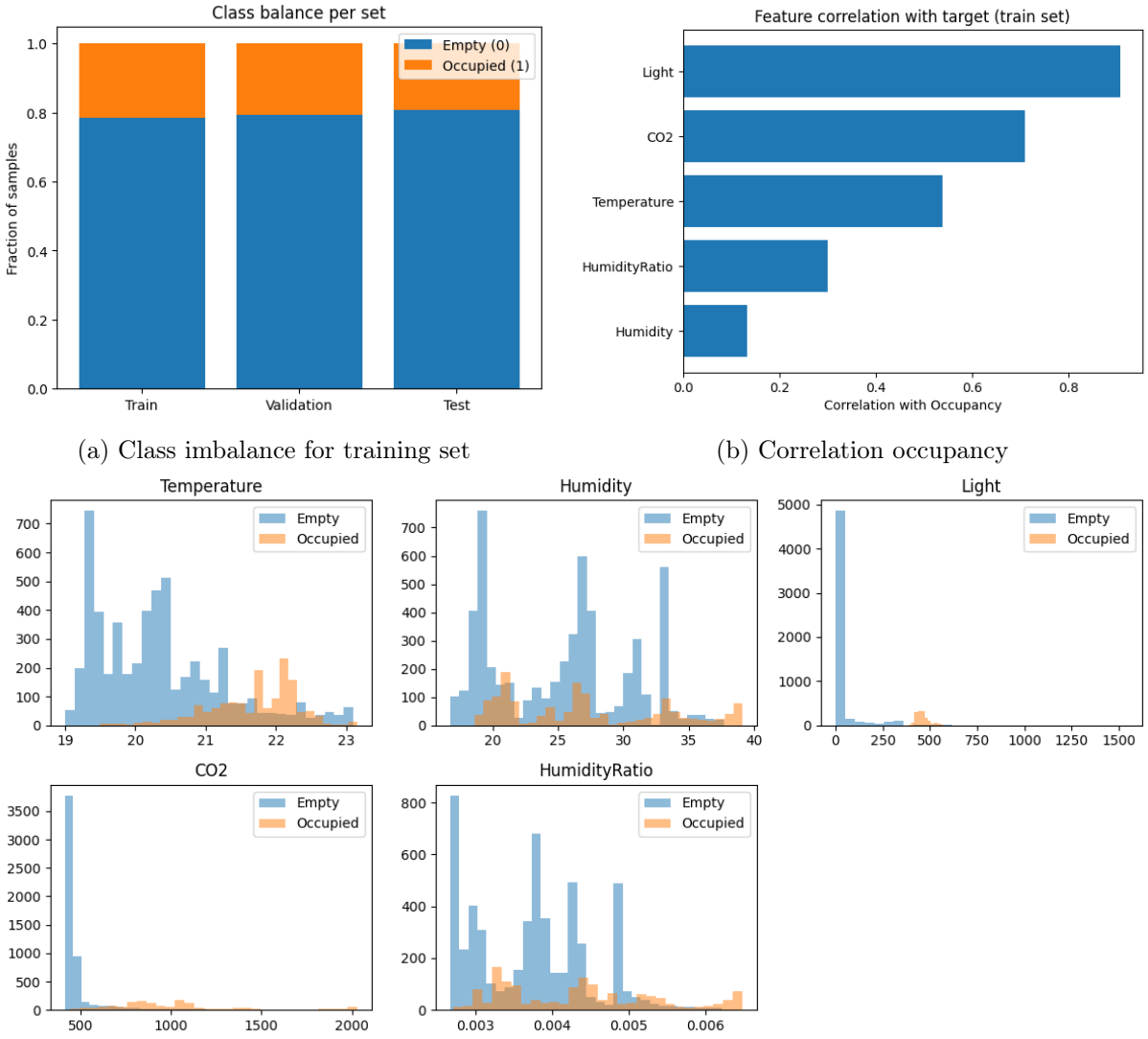
Given the first five quantities for a new record, our task is to predict the sixth: whether or not the room was occupied.

2 Dataset analysis

The dataset has already been divided into a training set of 6943 records, a validation set of 400 records, and a test set of 800 records. Each set is stored as a text file, with one record for each row, and with the features separated by spaces. By analyzing the .txt files from the three different datasets—training, test, and validation—it can be seen that the provided data has the following characteristics:

- **Class imbalance**. Like shown in Figure 1a, all three splits are imbalanced toward "unoccupied": ∼78.5% class 0 in train, ∼79.3% in validation, ∼80.6% in test. The splits are consistent with each other, so this isn't a sampling artifact — it reflects real occupancy patterns (rooms are empty more often than not).
- **Feature scales differ wildly**. Like shown in Figure 1c, light ranges 0–∼1546 Lux, CO2 ranges ∼410–2030 ppm, Temperature is a narrow 19–23°C band, and HumidityRatio is on the order of 0.003–0.006. This means feature standardization/normalization is worth doing before training (visible in the notebook), especially for models sensitive to scale like logistic regression trained with gradient descent.

- **Light and CO2 are by far the strongest predictors**. Correlation with Occupancy: Light ≈ 0.91, CO2 ≈ 0.71, Temperature ≈ 0.54, HumidityRatio ≈ 0.30, Humidity ≈ 0.13. This makes physical sense: lights are usually switched on when someone's in the room, and CO2 rises with human presence/breathing. We note that Light alone is almost a proxy for occupancy. This correlation is visible in Figure 1b.
- **No missing values**
- **Splits are IID-consistent**. Feature means/stds are close across train/validation/test, meaning the split looks like a proper random (or at least representative) partition.



(a) Class imbalance for training set

(b) Correlation occupancy

(c) Correlation occupancy

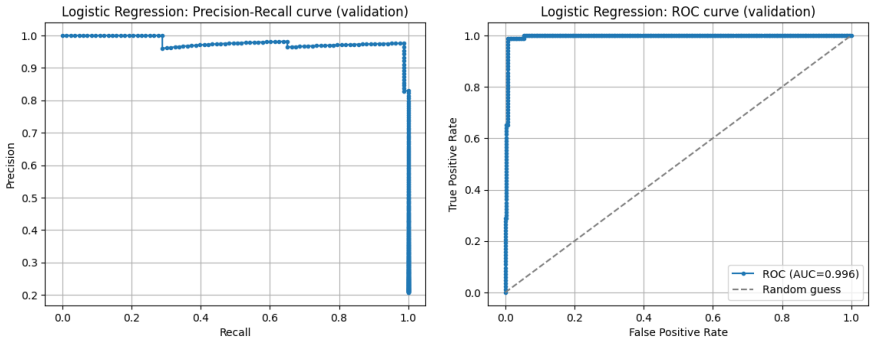
Figure 1: Dataset analysis

3 Models selection

3.1 Logistic Regression

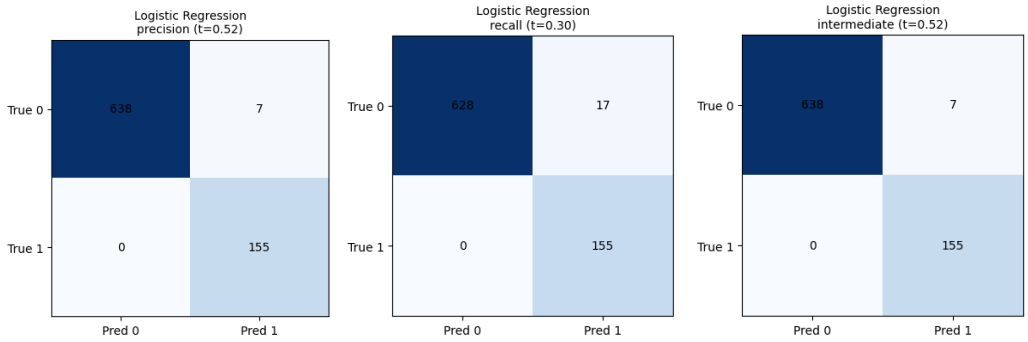
We trained logistic regression via gradient descent on standardized features, minimizing binary cross-entropy with an optional L_1 or L_2 penalty choosing the regularization strength λ via a

grid search over $\lambda \in \{10^{-4}, \ldots, 1\}$ for both variants (10,000 steps each), selecting the value that maximizes validation accuracy. As expected, during the grid-search L_1 collapses sharply at high λ (sparse solutions) while L_2 degrades more gradually, as expected from theory.
After training, we then swept the decision threshold on the validation set to build the Precision-Recall and ROC curves (Figures 2a, 2b) and selected three operating points: precision-oriented (best precision with recall ≥ 0.90), recall-oriented (best recall with precision ≥ 0.90), and intermediate (F_1-maximizing). The precision-oriented and intermediate thresholds coincide at $t = 0.52$, where the model still reaches perfect test recall (1.00) at 0.957 precision — a sign of well-calibrated, finely tunable probabilities. Figures 3a–3c show the corresponding confusion matrices.



(a) Precision/recall curve (b) ROC curve

Figure 2: Logistic Regression curves



(a) Precision-oriented (b) Recall-oriented (c) Intermediate version

Figure 3: LR models confusion matrices

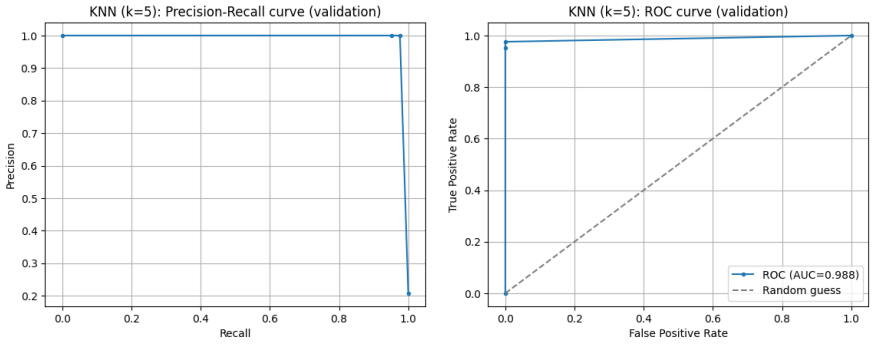
3.2 Linear SVM

The linear SVM minimizes the hinge loss with an L_2 weight-decay penalty (standard soft-margin formulation), on the same standardized features. As with logistic regression, $\lambda \in \{10^{-4}, \ldots, 1\}$ was selected via a validation-accuracy grid search (10,000 steps per value).

The same threshold-sweeping procedure, applied to the sigmoid-mapped decision margin, gives the three operating points; the corresponding curves closely resemble logistic regression's and are omitted for brevity. Of the four models, the linear SVM has the lowest test accuracy (0.988 at the intermediate point, vs. 0.991–0.993 for the others) — consistent with its hinge-loss margin being an uncalibrated score rather than a genuine probability, unlike logistic regression's cross-entropy-trained output.

3.3 K-Nearest Neighbors

KNN classifies each point by majority vote among its k nearest neighbors in the standardized feature space; k was chosen via a validation-accuracy grid search over a range of representative values, selecting $k = 5$. With $k = 5$, the estimated probability of occupancy can only take 6 distinct values $(0, \frac{1}{5}, \ldots, 1$ — the fraction of neighbors in the positive class), so the threshold sweep behind Figures 4a and 4b has only a handful of candidate thresholds instead of a near-continuous range. This is why all three operating points coincide at probability threshold $t = 0.40$: with so few achievable (precision, recall) pairs, the same threshold happens to be optimal under every criterion; this is visible in Figure 5.



(a) Precision/recall curve (b) ROC curve

Figure 4: 5-Nearest Neighbors curves

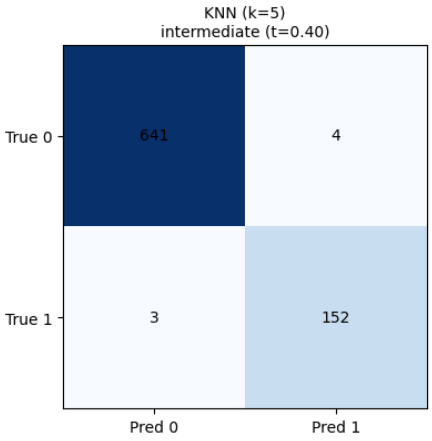


Figure 5: Confusion matrix

3.4 MLP Neural Network

Given how strongly Light and CO2 already correlate with occupancy, and how well the simpler models perform, the MLP was introduced mainly to test whether non-linearity could improve on them, not because it was expected to be necessary. We used a small feed-forward network, $5 \to 16 \to 8 \to 1$ with ReLU activations (Figure 6), outputting a raw logit trained with `BCEWithLogitsLoss` for numerical stability.

With the architecture fixed, the L_2 weight decay of the Adam optimizer was chosen via a validation-accuracy grid search. The resulting model is the best of the four (accuracy 0.993, precision/recall 0.981 at the intermediate point), though only marginally ahead of KNN and logistic regression — consistent with the dataset being close to linearly separable, leaving little room for a more expressive model to add value.

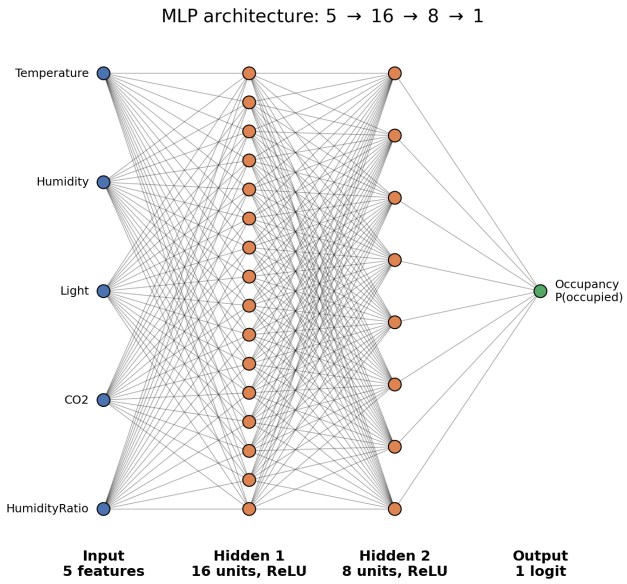


Figure 6: MLP architecture: 5 input features, two ReLU hidden layers (16 and 8 units), one output logit.

4 Models evaluation and conclusion

For every model, the three operating points (precision-oriented, recall-oriented and intermediate) selected on the validation set were applied, unchanged, to the test set. Table 1 reports the resulting metrics.

Model	Operating point	Threshold	Accuracy	Precision	Recall
Logistic Regression	precision	0.52	0.9912	0.9568	**1.0000**
	recall	0.30	0.9787	0.9012	**1.0000**
	intermediate	0.52	0.9912	0.9568	**1.0000**
Linear SVM	precision	0.62	0.9875	0.9560	0.9806
	recall	0.44	0.9750	0.8857	**1.0000**
	intermediate	0.62	0.9875	0.9560	0.9806
KNN (k=5)	precision	0.40	0.9912	0.9744	0.9806
	recall	0.40	0.9912	0.9744	0.9806
	intermediate	0.40	0.9912	0.9744	0.9806
MLP	precision	0.52	**0.9925**	**0.9806**	0.9806
	recall	0.00	0.9800	0.9064	**1.0000**
	intermediate	0.52	**0.9925**	**0.9806**	0.9806

Table 1: Test-set performance at the three selected operating points, for all four models. Bold marks the best value in each column.

Overall, the MLP achieves the highest accuracy and precision overall (0.9925 and 0.9806 respectively, at its precision-oriented and intermediate operating points), confirming it as the strongest model on this task, if only by a small margin over KNN and logistic regression. Perfect recall (1.0) is reached by five different operating points across three separate models — logistic regression at all three of its operating points, the linear SVM's recall-oriented point, and the MLP's recall-oriented point — showing that catching every occupied instance is comparatively

easy to achieve on this dataset, typically at a moderate cost in precision; the linear SVM is consistently the weakest of the four (it never reaches either the best accuracy or the best precision). The gap between the best and worst model is small, suggesting the dataset is close to linearly separable — consistent with the very high correlation of Light and CO2 with occupancy observed during the initial data analysis.

I affirm that this report is the result of my own work and that I did not share any part of it with anyone else except the teacher.